Math 261-001
Quiz 6, October 9

Name	CSU ID #

Choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. Use spherical coordinates to evaluate

$$I = \iiint_E \frac{dV}{(x^2+y^2+z^2)^{3/2}},$$

where E is the part of the shell $1 \le x^2+y^2+z^2 \le 4$ satisfying $z \ge \sqrt{x^2+y^2}$.

2. A fly follows the path C given by

$$\vec{r}(t) = (\cos t, \sin t, 2+\cos t+\sin t), \qquad 0 \le t \le 2\pi.$$

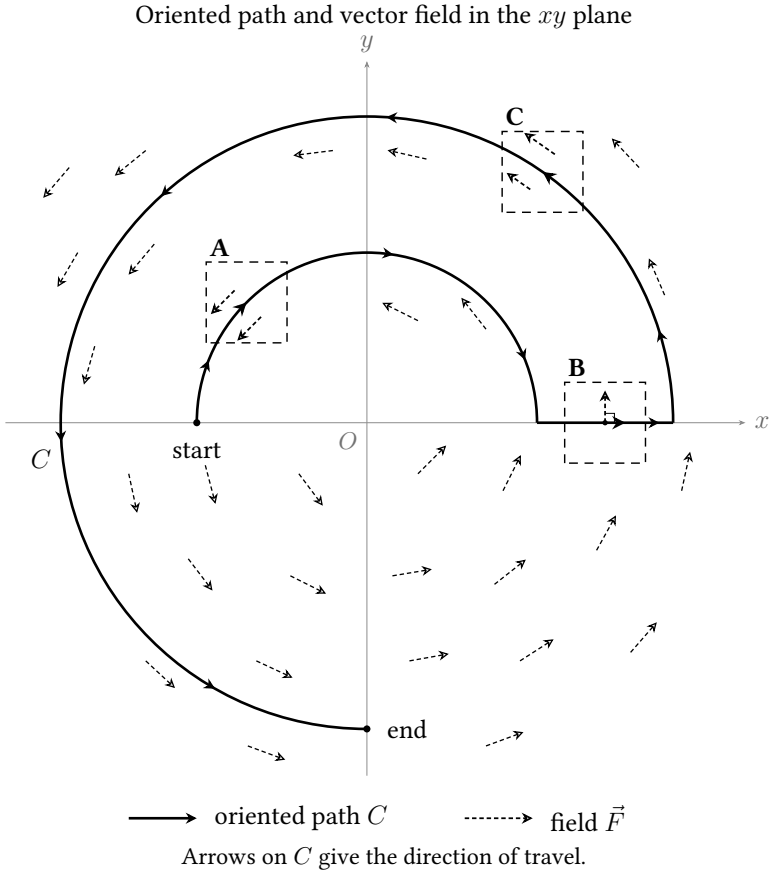
Along the path, there's a scent with intensity $\sigma(x,y,z) = \sqrt{1+(x-y)^2}$. Compute its scent-weighted flight length $S = \displaystyle\int_C \sigma\, ds$.

3. Compute $\displaystyle\int_C \vec{F} \cdot d\vec{r}$, where

$$\vec{F}(x,y) = \left(\frac{y\cos x}{x^2}, -\frac{\sin x}{2x}\right), \qquad \vec{r}(t) = (t, t^2), \quad \frac{\pi}{2} \le t \le \pi,$$

and C is traversed in the direction of increasing t.

4. For the oriented piece of C inside each dashed square A, B, and C, is the contribution to $\int_C \vec{F} \cdot d\vec{r}$ **positive**, **negative**, or **zero**? Use the direction arrows on C and the dashed field arrows. The indicated alignment holds along each highlighted piece.



A:

B:

C: